
plan-failure-bench: A Machine-Checkable Benchmark of How Language Model Planners Fail

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<https://github.com/munawarkazmi/plan-failure-bench>

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Abstract

Planner evaluations for language models usually report a success rate, rely on labels written by hand, and often score with a human or LLM judge. We propose construction rules that remove all three and demonstrate them in plan-failure-bench. Each of 60 instructions over two symbolic indoor environments either admits a valid plan or plants one trap from six families (unreachable goal, missing capability, ambiguous referent, precondition, sequencing, constraint), and each label carries a proof that continuous integration re-checks on every commit. Models answer in a small JSON action language with explicit infeasible and clarify responses, so a deterministic checker scores every answer and every detection count is reported with its false positive count on feasible instructions. The checker is differentially tested against an independent PDDL toolchain, and every instruction also appears in a semantically obfuscated form produced by a versioned, invertible renaming. At this scale the results on four models are hypotheses. The models fail in different ways; the obfuscation versioning exposed one of our own apparent findings as a token artefact; no model distinguishes a missing capability from an unreachable goal; and the strongest model tested, Gemini 3.6 Flash, nearly clears the suite in both conditions, so the failure findings apply to smaller models. This is a working paper; every number in it is reconstructible from the repository.

1 Introduction

When a household robot is driven by a language model planner, the most dangerous failures are confident ones: the model that plans a route through a room it was told never to enter, or fetches one of two identical cups without asking which. Before asking whether a language model plans well, a benchmark should ask whether the model notices when planning is the wrong response altogether.

plan-failure-bench makes that question checkable by machine. The contribution is a way of constructing planner evaluations: every scoring decision is mechanical, every ground-truth label carries a proof, and every number in this paper is generated from committed run records. The grid of four models and two environments reported here demonstrates the method at small scale (30 instructions per environment, models chosen by free-tier access), so its findings are stated as hypotheses. The construction rules ensure that those hypotheses cannot come from a judge, a mislabelled instruction, or a transcription error, and Section 6 describes two artefacts the method caught in its own first obfuscation scheme. The rules:

1. **Finite symbolic worlds.** States are sets of ground atoms with no numerics, so every label can be proved mechanically.
2. **Proof obligations in continuous integration.** Every seed’s label carries a mechanical proof (a validating reference plan, an unreachability proof from a sound abstraction, a capability grant that flips infeasibility, or exhaustive referent bindings), re-proved on every commit by 589 tests.
3. **Refusal and clarification as first-class answers.** The response language contains `infeasible(reason)` and `clarify(candidates)` terminals, so detection is scored mechanically and always reported alongside the paired false positive count on feasible instructions.

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4. **A semantically obfuscated condition.** Content words are renamed by a versioned bijection applied to the prompt and inverted on the response, separating state tracking from lexical pattern matching while the checker scores only the canonical world.
 5. **Two structurally contrasting environments.** The second environment varies topology and constraint structure specifically to test whether findings from the first are artefacts of one map.
 6. **Open, replayable records, and no transcribed numbers.** Every run is committed as one JSON record per seed, both scoring policies re-score offline without model calls, model names resolve through a committed manifest of exact API checkpoints, and the results tables in this paper are generated from the records.

2 Related work

Evaluating LLM planning. PlanBench established systematic evaluation of LLM planning over symbolic domains and its successors have repeatedly found generation weak (Valmeekam et al., 2023a;b). The same line introduced Mystery Blocksworld, obfuscating a familiar domain to separate reasoning from lexical recall, and found that plan generation collapses; plan-failure-bench inherits that manipulation, applies it as a versioned bijection, and asks which *judgements* survive semantic removal as well as whether generation does. Unsolvability recognition itself is not new: PlanBench ships unsolvable instances scored by true and false negative rates, and Valmeekam et al. (2024) report o1-preview explicitly recognising 27% of unsolvable Blocksworld instances, 16% under randomised obfuscation, while wrongly declaring 11.5% of solvable obfuscated instances impossible. What plan-failure-bench adds is breadth and pairing: six trap families under one response protocol, detection always paired with false positives, and per-seed proof obligations. Pipeline alternatives such as LLM+P hand the search problem to a classical planner and use the model only for formalisation (Liu et al., 2023); plan-failure-bench instead measures the model as the planner, because the failure modes of that configuration are what the benchmark exists to characterise.

Feasibility, ambiguity, and safety as separate literatures. Each trap family here has a single-family predecessor. Plancraft includes intentionally unsolvable crafting tasks scored mechanically: given an explicit impossible action, its best model reaches F1 0.71 at declaring impossibility, and models become overeager, trading success on solvable tasks for refusals (Dagan et al., 2025). AmbiK pairs every ambiguous kitchen instruction with an unambiguous counterpart and scores whether detection methods can tell the two apart (Ivanova et al., 2025), and the asking-for-help line, KnowNo’s calibrated uncertainty and CLARA’s command classification, trades help requests against success guarantees (Ren et al., 2023; Park et al., 2024). Two 2026 benchmarks make the pairing suite-wide in the tool-use setting: TREK pairs 533 feasible travel-planning tasks with 267 provably infeasible ones whose causes are typed (route, entity, budget), scores with deterministic rules and no LLM judge, and grades a bare refusal strictly below one that names the right cause (Qi et al., 2026); AgentAbstain pairs every should-abstain task with a should-act variant differing in exactly one perturbation across eight scenarios, so that no constant policy can exceed 50% paired accuracy, though an LLM judge scores the terminal response (Liu et al., 2026). Paired or trade-off measurement therefore exists within single families and, in tool-use domains, suite-wide, including the false impossibility claims measured by Valmeekam et al. (2024); plan-failure-bench carries the pairing into embodied planning across six trap families under one protocol. SafeAgentBench shows embodied agents complying with hazardous instructions and documents the over-refusal trade-off directly, a safety module raising rejection of safe tasks alongside hazardous ones (Yin et al., 2024); HomeBench evaluates valid and invalid instructions in smart home device control, where even frontier models fail badly on invalid multi-device cases (Li et al., 2025). XSTest documents the chat-model analogue of over-refusal, driven by surface features of safe prompts (Röttger et al., 2024), and Semantic Denial of Service attacks show robot refusal triggering on injected surface-level safety phrases regardless of physical state (Steinberg & Gal, 2026), adversarial evidence of the same anchoring our obfuscation manipulation measures. The nearest neighbour to our aims is the Yes-Man Syndrome benchmark (Yeke et al., 2026), which measures abstention by embodied agents over ambiguous, physically infeasible, false-premise, and sensing-unresolvable instructions. Its instructions are generated by deterministic constraint derivation, but its scoring uses an LLM judge, it measures only whether agents abstain when they should, with no false positive column for valid instructions, and it has no semantic obfuscation condition; its categories all target abstention,

while half of plan-failure-bench’s taxonomy consists of feasible instructions carrying planted execution traps, which is what makes the planted versus observed confusion matrix possible. To our knowledge no prior benchmark manipulates surface semantics while measuring refusal, the manipulation behind our finding that refusal behaviour is anchored to surface semantics in opposite directions for different models.

Error taxonomies and ground truth. The Embodied Agent Interface catalogues fine-grained error types across goal interpretation, decomposition, sequencing, and transition modelling (Li et al., 2024). plan-failure-bench differs in direction: instead of classifying observed errors after the fact, it plants exactly one labelled trap per instruction and reports the confusion matrix between planted and observed, asking whether models fail as predicted. That requires ground truth that can be checked independently, which places this work in the plan-validation tradition of VAL (Howey et al., 2004): verdicts come from a deterministic checker differentially tested against an independent PDDL executor, pyperplan (Alkhazraji et al., 2020), labels are proofs re-run in continuous integration, and no human or model judgement appears anywhere in the scoring loop. Plancraft also scores mechanically inside its crafting environment, TREK’s evaluator is deterministic rules over its travel knowledge base, and SIMMER validates cooking plans with a state-machine executor over a curated symbolic kitchen world; its latent failures, flawed plans that execute without immediate feedback, are single-domain kin to our silent-violation constraint seeds, though all of its tasks are solvable, so refusal, paired false positives, and semantic manipulation are out of its scope (Lu et al., 2026). RoboBench, by contrast, delegates plan feasibility scoring to a multimodal model acting as a world simulator (Luo et al., 2025) and the Yes-Man benchmark scores with an LLM judge. What no benchmark above combines is judge-free scoring with per-seed proof obligations re-run in continuous integration and differential testing of the checker itself.

Proving that no plan exists. Our infeasibility labels rest on a standard result in classical planning: if a projection of a task onto a subset of its variables has no solution, neither does the task. Bäckström et al. (2013) search such projections to detect unsolvable instances, and Hoffmann et al. (2014) design merge-and-shrink abstractions for the same purpose. Our abstraction is simpler: for each goal literal it tracks the robot, the doors, and one item exactly and over-approximates every other item, which is enough to prove every infeasible label in both suites. Eriksson et al. (2017) note that plans are routinely checked by independent validators while claims of unsolvability were not, and propose certificates that a separate verifier can check. Our unreachable proofs are checked only by our own code; emitting such certificates would give them the independent check our plans receive.

Measurement validity and judge reliability. The construction rules in the introduction respond to documented pathologies of current evaluation practice. A systematic review of 445 LLM benchmarks found nearly every one weak on at least one construct validity criterion, the operationalisation of abstract phenomena often insufficient, and issued eight recommendations including defining the phenomenon, preparing for contamination, and justifying construct validity (Bean et al., 2025); plan-failure-bench answers the narrower internal-validity question by proof, since what each instruction measures is its planted, machine-proved label. On judges, a systematic evaluation of 21 LLM judges over roughly 541,000 judgments finds that raw exact-match agreement overstates discriminative ability (Cohen’s kappa 33.8 to 41.3 percentage points lower on MT-Bench), that judge rankings shift by up to 14 positions across benchmarks, and that production-deployed judges combine near-perfect test-retest reliability with severe position bias (Norman et al., 2026); plan-failure-bench scores with no judge anywhere, so those pathologies do not apply. Shallow-cue exploitation has also been measured directly: simple classifiers over answer n-grams score highly on several multiple-choice benchmarks while lacking the capability being tested (Pacchiardi et al., 2024). The obfuscated condition answers the same concern by construction, removing the surface cues while provably preserving task structure.

3 Benchmark design

3.1 World model

An environment is a set of rooms connected by doors (open, closed, or locked), items with categories and properties located in rooms, a robot with a capability profile over a fixed action vocabulary (goto, open, close, pick, place, and optionally unlock), a single-slot gripper, and trajectory invariants that must hold in

every state along a plan. Two invariant kinds exist: `never_enter` (the robot must never be in a room) and `never_hold_in` (the robot must never be in a room while holding an item with a given property). Invariant text is shown to the model verbatim, so the model is told every constraint it is scored against. State is a finite set of ground atoms over a fixed object universe with no numerics, so the state space is finite.

3.2 Task language

Models receive a fixed rendering of the environment and one natural language instruction, and must answer with exactly one JSON object: a `plan` (a list of actions), `infeasible` with a reason from `{unreachable, missing_capability, constraint}`, or `clarify` with candidate referents. The strict scoring policy, the headline, requires exactly a response-shaped JSON object; a documented lenient policy re-scores stored records offline by extracting the first response-shaped JSON object, separating format discipline from planning ability.

3.3 Trap taxonomy and proof obligations

We call each labelled instruction a *seed*, and the 30 seeds of one environment a *suite*. Each seed either admits a valid plan or plants exactly one trap. Labels carry these proof obligations, re-proved on every test run:

- **valid:** a reference plan validates under the checker and under independent PDDL execution, and no infeasibility proof exists.
- **unreachable_goal:** provably unreachable under a sound over-approximating abstraction, and still unreachable with the full action vocabulary granted.
- **missing_capability:** provably unreachable under the robot’s profile, not provable once a named capability is granted, and a capability reference plan validates in the augmented environment.
- **ambiguous_referent:** at least two same-category bindings, each with a validating reference plan and no infeasibility proof; the expected answer is `clarify` with exactly the category’s members.
- **precondition_trap** and **sequencing_trap:** the goal is feasible (reference plan validates); a decoy plan documenting the intended error produces exactly the declared trap verdict.
- **constraint_trap:** either feasible with a decoy that executes fully, achieves the goal, and silently breaches an invariant, or provably infeasible solely because of an invariant.

A *decoy plan* is the tempting wrong plan a trap is built around. Precondition and sequencing traps together are the *ordering traps*, seven per suite. Four seeds recur by name in the results. The *silent-violation* seed is the feasible constraint seed, whose decoy executes fully and reaches the goal while breaching an invariant; the *compound* constraint seed asks for two items, only one of which may be carried to the target; two missing-capability seeds are *locked-door* seeds, sealed behind a door this robot cannot unlock; and the third is the *inexpressible-verb* seed, which asks for an action no robot has, such as mopping.

A trap verdict on a feasible seed shows that the model missed the trap. It does not show that the model took the decoy: `precondition_violation` is the verdict for any plan that fails early, whatever the cause, so an observed `precondition_violation` on a precondition trap is consistent with the decoy but does not identify it.

3.4 Environments

house_01: seven rooms, six doors, ten items, two `never_hold_in` invariants (each property carried by exactly one item), a doorless cellar whose isolation the environment description states outright, and a locked store room behind the withheld unlock capability. It contains one cycle (kitchen, hallway, living room), passable only through a closed door.

office_01: nine rooms, eight doors, eleven items, built to contrast with `house_01` in structure. A five-room ring reachable through open doors makes route choice pervasive. A `never_enter` room sits on the ring, so the short route between reachable rooms can silently violate an invariant by movement alone. A `never_hold_in`

property is carried by three items rather than one. A two-room annex is disconnected from the rest of the map, but both its rooms have doors, so no “has no doors” statement appears and the isolation must be inferred from the connection list. Ambiguous referent candidates can sit in different rooms, and one decoy traps the single-slot gripper itself. Both suites plant the same label distribution (9 valid, 4 unreachable, 3 missing capability, 3 ambiguous, 4 precondition, 3 sequencing, 4 constraint), keeping confusion matrix columns comparable across environments.

3.5 Obfuscated condition

Semantic content words (rooms, doors, items, categories, properties, actions, door states) are replaced by deterministic nonsense tokens; structural vocabulary (room, door, item, gripper) and the response protocol are retained. The renaming is a bijection applied to the assembled prompt and inverted on the model’s raw response before checking, so the checker, the seeds, and every label proof operate only on the canonical world and no second world model exists to drift. Version 1 tokens were confusable enough that model copy errors were indistinguishable from hallucination; version 2 tokens guarantee unique opening syllables and pairwise edit distance of at least three. Every record stores the obfuscation version it was produced under.

4 Ground truth guarantees

The checker simulates each plan step by step and assigns exactly one verdict: `valid`, `precondition_violation`, `constraint_violation`, `goal_not_achieved`, `hallucinated_entity`, `unavailable_action`, or `malformed`, plus the two terminals. It is differentially tested against pyperplan over hand-written trap plans and hundreds of seeded random and guided plans, with first-failing-step agreement required. A second compilation turns both invariant kinds into STRIPS preconditions, the standard compilation of PDDL 3 *always* constraints over state atoms, and there the first inapplicable step must equal the checker’s first breach step, so the oracle covers constraint verdicts as well as executability and goal satisfaction. Infeasibility labels are proofs from a per-literal over-approximating abstraction whose direction is sound: every real trajectory has an abstract counterpart, so goals the abstraction cannot reach are truly unreachable, and reachable-looking goals are proved feasible constructively by reference plans, never by the abstraction.

5 Experimental setup

Four models, all under a fixed prompt at temperature 0 with one decode per seed, each covering both suites in both conditions: Llama 3.3 70B (Groq API), Qwen 2.5 7B Instruct (run locally through Ollama as the 4-bit Q4_K_M quantization of the `qwen2.5:7b-instruct` tag), Gemini 3.1 Flash Lite, and Gemini 3.6 Flash (both Gemini API; the latter a reasoning model). Each request carried the prompt as one user message, temperature 0, and the output limit in Table 1, and nothing else. No JSON mode or structured output constraint was used, so format failures measure the models’ unconstrained output. No reasoning or thinking parameter was sent, so both Gemini models ran with the provider’s default thinking, and their reasoning tokens are not recorded. Temperature 0 does not make a hosted API replay identically, so each single decode is one sample of the model’s behaviour. Seven of the 1350 committed responses stop at the output limit mid-answer, all malformed under the strict policy (`tools/audit_truncation.py` lists them). One is in a superseded version 1 run. Of the six in the current grid, five are Llama office_01 responses from which lenient extraction recovers an earlier plan, and one is a Flash Lite plan that moves between two rooms 483 times before the limit cuts it off. Table 1 records the exact API checkpoint behind each run alias; every committed record carries the alias, and records written from now on carry the checkpoint string itself, so the mapping from a table row to the model that produced it is recorded in the repository. The superseded version 1 obfuscated runs of Llama and Qwen are retained and marked in the tables, because the artefacts they exposed are part of the methodology’s history.

Model	Run alias in records	API checkpoint	Output limit
Llama 3.3 70B	groq_llama70b	llama-3.3-70b-versatile	2000
Qwen 2.5 7B (Q4_K_M)	local_qwen	qwen2.5:7b-instruct	2000
Gemini 3.1 Flash Lite	gemini_flash_lite	models/gemini-3.1-flash-lite	8000
Gemini 3.6 Flash	gemini_flash	models/gemini-3.6-flash	8000

Table 1: Model aliases in the committed records, the exact API checkpoints they called, and the output token limit of every request, generated from the committed manifest.

Model	Environment	Condition	Note	Format failures	Traps detected	Exact reasons	False positives	Valid solved
Llama 3.3 70B	house_01	plain		18/30	9/13	6	3/17	5/9
Llama 3.3 70B	house_01	obfuscated (v1)	superseded	23/30	11/13	9	1/17	1/9
Llama 3.3 70B	house_01	obfuscated (v2)		26/30	10/13	8	0/17	5/9
Qwen 2.5 7B	house_01	plain		3/30	2/13	0	0/17	2/9
Qwen 2.5 7B	house_01	obfuscated (v1)	superseded	5/30	3/13	1	0/17	1/9
Qwen 2.5 7B	house_01	obfuscated (v2)		10/30	3/13	1	3/17	1/9
Gemini 3.1 Flash Lite	house_01	plain		0/30	12/13	8	4/17	6/9
Gemini 3.1 Flash Lite	house_01	obfuscated (v2)		0/30	7/13	6	1/17	5/9
Gemini 3.6 Flash	house_01	plain		0/30	13/13	10	0/17	9/9
Gemini 3.6 Flash	house_01	obfuscated (v2)		0/30	13/13	10	0/17	9/9
Llama 3.3 70B	office_01	plain		24/30	9/13	5	1/17	2/9
Llama 3.3 70B	office_01	obfuscated (v2)		27/30	7/13	5	1/17	2/9
Qwen 2.5 7B	office_01	plain		4/30	1/13	0	0/17	2/9
Qwen 2.5 7B	office_01	obfuscated (v2)		13/30	3/13	2	2/17	3/9
Gemini 3.1 Flash Lite	office_01	plain		1/30	10/13	6	7/17	4/9
Gemini 3.1 Flash Lite	office_01	obfuscated (v2)		1/30	5/13	4	2/17	2/9
Gemini 3.6 Flash	office_01	plain		0/30	13/13	10	0/17	9/9
Gemini 3.6 Flash	office_01	obfuscated (v2)		0/30	13/13	10	0/17	9/9

Table 2: Every complete committed run. Format failures are strict-policy malformed responses out of 30; all other columns use lenient extraction. Traps detected counts the 13 seeds per suite whose expected answer is a terminal, and should be read with the paired false positives on the 17 feasible instructions. Exact reasons counts detections whose reason code or candidate set was exactly right.

6 Results

Numbers below are counts under the lenient extraction policy, one decode per seed at temperature 0 unless a paragraph says otherwise, with strict format failures reported per run. At $n = 30$ per condition they are hypotheses. Table 2 is generated directly from the committed records.

6.1 Failure profiles

Each model fails in its own way. Llama detects most traps but breaks the response format in 18 of 30 plain responses by wrapping correct JSON in prose. Qwen keeps the format and almost never refuses: zero false positives in plain on both environments, with nearly every trap ending in a precondition violation. Gemini 3.1 Flash Lite keeps the format on house_01 and detects 12 of 13 traps there, but solves none of house_01’s seven ordering traps and refuses more feasible instructions than any other model. Each profile holds in direction on office_01. Llama’s plain office_01 run detects exactly as many traps as on house_01 (9 of 13), while its format failures rise to 24 of 30 and its valid-seed success falls to 2 of 9.

The strongest model tested nearly clears the suite. Gemini 3.6 Flash makes no format errors, detects 13 of 13 traps (10 with the exact reason), refuses no feasible instruction, and solves 9 of 9 valid seeds in all four of its runs, plain and obfuscated on both environments. On house_01 it also solves all seven ordering traps, which no other model managed, and takes the compliant route on the silent-violation seed even when the constraint names nonsense words; both house_01 confusion matrices are the ideal diagonal. On office_01 it detects all four unreachable seeds in both conditions, including the two annex seeds whose isolation must

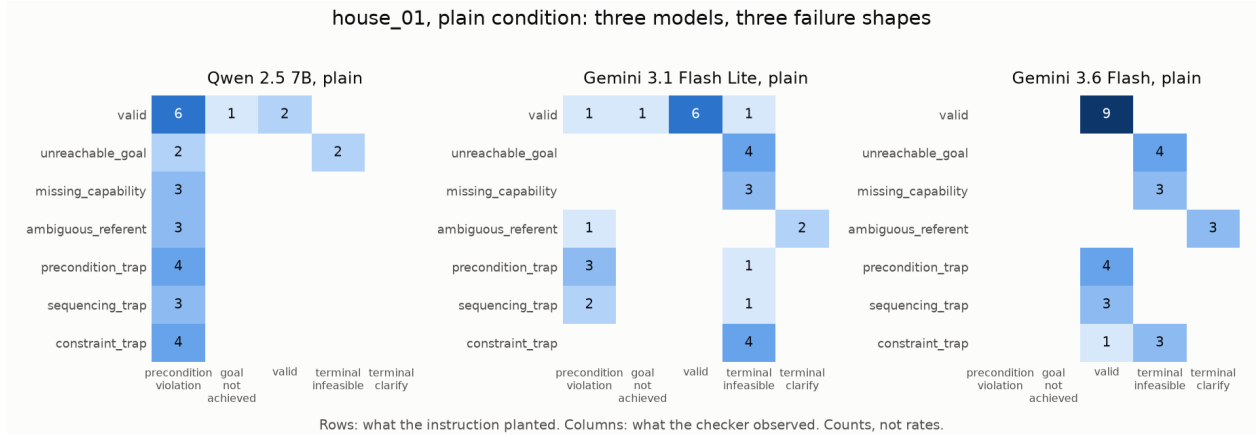


Figure 1: Planted trap (rows) against observed verdict (columns) for three of the eighteen runs. The same 30 seeds produce three shapes: Qwen sends almost every trap to one execution-failure column, Flash Lite refuses across the matrix including on feasible seeds, and Gemini 3.6 Flash returns the diagonal. Full grids for every run are in Appendix A.

be inferred from the connection list. Its one blemish is an office_01 sequencing trap whose stated order closes the robot’s own route: in both conditions it returns a plan that satisfies one of two goal conjuncts (**goal_not_achieved**). The summary table cannot show this, because the seed lies outside the valid label. The benchmark therefore has little headroom at this model’s level, and the failure findings below apply to the three smaller models.

No model separates a missing capability from an unreachable goal. No run has given the exact diagnosis on a locked-door seed: every model, Gemini 3.6 Flash included, calls them unreachable. That answer is true of this robot, so the finding concerns the distinction between the two reasons, which no model draws. The only exact missing-capability reasons came from Gemini 3.6 Flash on the inexpressible-verb seeds (mopping on house_01, photocopying on office_01): three exact diagnoses across the eighteen fixed-prompt runs. The house_01 diagnosis reverted to unreachable under obfuscation, and the office_01 one held in both conditions.

6.2 Obfuscation

Our first token scheme produced two artefacts, and versioning caught both. Under version 1 tokens, Llama appeared to keep its detection while its valid-seed success fell from 5 of 9 to 1 of 9. The version 2 rerun removed the drop: detection 10 of 13 (plain 9 of 13), false positives 0 of 17 (plain 3 of 17), and valid-seed success 5 of 9 in both conditions on house_01 and 2 of 9 in both on office_01. The same scheme produced 15 hallucinated-entity verdicts for Qwen on house_01 (version 2: 1) and 4 for Llama (version 2: 0; Table 3); these were confusable tokens copied wrongly. Records carry their obfuscation version, and the superseded runs stay in Table 2. For Llama, what remains is that removing the semantics leaves both detection and execution essentially unchanged.

Over-refusal moves with surface semantics. Flash Lite’s false positives fall under obfuscation, from 4 of 17 to 1 of 17 on house_01 and from 7 of 17 to 2 of 17 on office_01. Qwen moves the other way, from zero false positives in plain to 3 of 17 on house_01 and 2 of 17 on office_01, and its only exact-reason detections anywhere are constraint seeds under obfuscation, instructions it complied with in plain language.

For Flash Lite, stated isolation survives obfuscation and inferred isolation does not. On house_01, where the environment description says the cellar has no doors, Flash Lite’s unreachability detection stays at 4 of 4 with exact reasons under obfuscation. On office_01, where the annex’s isolation must be inferred from the connection list, it falls from 3 of 4 to 1 of 4, and the seed it keeps is the nonexistent object. Gemini 3.6

Flash detects all four office_01 unreachable seeds in both conditions, so this split separates the two Gemini models.

6.3 Stability across samples and prompts

Qwen’s profile survives sampling. Five decodes per seed at temperature 0.7 (plain, house_01) gave the same lenient verdict on 26 of 30 seeds. Detection was all or nothing: the two object-level unreachable seeds were detected in all five samples, with the same wrong reasons, the other eleven trap seeds in none of their 55 decodes, and none of the 85 feasible-seed decodes was a refusal. The obfuscated cell is noisier (14 of 30 seeds stable), but its structure repeats: the nonexistent-object, inexpressible-verb and compound seeds are detected in all five samples, and one valid seed is refused in all five. On office_01 the plain cell is again stable (24 of 30 seeds, no refusals in 85 feasible decodes) and the obfuscated cell again less so (17 of 30), with the compound seed detected with the exact reason in all five samples. Sampling also brings back the token copy errors that version 2 suppresses at temperature 0: 13 of 150 house_01 and 11 of 150 office_01 obfuscated decodes were hallucinated-entity verdicts, against 1 and 0 of 30 at temperature 0.

Llama’s detection varies between samples. The same protocol for Llama (plain, house_01) gives 19 of 30 seeds the same lenient verdict in all five samples, with strict format failures of 17, 18, 17, 18 and 16 of 30 and lenient extraction leaving 1, 4, 1, 0 and 1 malformed. Seven of the 13 trap seeds are detected in all five samples, five in some, and one in none; two of the 17 feasible seeds are refused at least once, one in every sample. Three infeasible seeds mix `terminal_infeasible` decodes with `precondition_violation` decodes, detected in one sample and planned into in the next. The silent-violation seed splits two to three: two decodes take the decoy, which executes fully and breaches the constraint, and three refuse a feasible instruction.

No prompt wording fixes Llama’s format failures. Over the same seeds (Llama, plain, house_01), the canonical prompt gives 18 of 30 strict format failures, a bare JSON-only variant 12 of 30, and a variant with the format rules moved to the end 15 of 30. The bare variant has a cost: 8 of its 30 responses contain no recoverable JSON, against 1 of 30 under the canonical prompt. The lenient planning metrics barely move across prompts (detection 8 to 10 of 13, false positives 2 to 3 of 17), which is the separation the two scoring policies exist to provide.

7 Conclusion

plan-failure-bench shows that a planner benchmark can be built without hand-written labels or judges: every label is proved, every response receives one mechanical verdict, and every detection count comes with its false positive count. Four results carry the most weight at this scale. The four models fail in distinct ways that the planted versus observed confusion matrix makes visible. Versioned obfuscation exposed one of our own apparent findings, Llama’s execution drop, as a token artefact. No model distinguishes a missing capability from an unreachable goal. And Gemini 3.6 Flash nearly clears the suite in both conditions, so the benchmark needs harder environments, longer plans, and more sampled cells before its failure findings can speak to the strongest models.

8 Limitations

The scope is four models (Llama 3.3 70B, Qwen 2.5 7B Instruct, Gemini 3.1 Flash Lite, Gemini 3.6 Flash), two environments, and 30 seeds per condition. Every table cell is one decode at temperature 0, so we report counts and attempt no statistical treatment yet. The $k = 5$ consistency experiment covers Qwen’s four cells (both environments, both conditions), where it found the single decodes representative in direction, and Llama’s plain house_01 cell; the remaining cells are unsampled. Two environments in one domain family (indoor rooms, doors, portable items) bound the generality of every claim. Prompt sensitivity is quantified for one model only (Llama, plain, house_01); the other models’ format behaviour under prompt variation is unmeasured. Qwen’s results describe its 4-bit quantized build, which may plan worse than the published

weights. The unreachability proofs are checked only by the benchmark’s own code. The lenient policy is a fixed syntactic extraction rule. All hosted models ran on free-tier quotas, which selected the model list.

9 Future work

Extending the $k = 5$ sampling protocol to the unsampled cells unlocks a statistical treatment and would most strengthen the claims here. More environments would test whether the taxonomy discriminates outside this domain family, and a second model at Gemini 3.6 Flash’s level would test whether its indifference to obfuscation holds for strong models generally.

References

- Yusra Alkhazraji, Matthias Frorath, Markus Grützner, Malte Helmert, Thomas Liebetraut, Robert Mattmüller, Manuela Ortlieb, Jendrik Seipp, Tobias Springenberg, Philip Stahl, and Jan Wülfing. Pyperplan, 2020.
- Christer Bäckström, Peter Jonsson, and Simon Ståhlberg. Fast detection of unsolvable planning instances using local consistency. In *Proceedings of the Sixth International Symposium on Combinatorial Search*, 2013.
- Andrew M. Bean et al. Measuring what matters: Construct validity in large language model benchmarks, 2025. arXiv preprint arXiv:2511.04703.
- Gautier Dagan, Frank Keller, and Alex Lascarides. Plancraft: An evaluation dataset for planning with LLM agents. In *Conference on Language Modeling*, 2025.
- Salomé Eriksson, Gabriele Röger, and Malte Helmert. Unsolvability certificates for classical planning. In *Proceedings of the Twenty-Seventh International Conference on Automated Planning and Scheduling*, 2017.
- Jörg Hoffmann, Peter Kissmann, and Álvaro Torralba. “distance”? Who cares? Tailoring merge-and-shrink heuristics to detect unsolvability. In *Proceedings of the 21st European Conference on Artificial Intelligence*, 2014.
- Richard Howey, Derek Long, and Maria Fox. VAL: Automatic plan validation, continuous effects and mixed initiative planning using PDDL. In *IEEE International Conference on Tools with Artificial Intelligence*, 2004.
- Anastasiia Ivanova, Eva Bakaeva, Zoya Volovikova, Alexey K. Kovalev, and Aleksandr I. Panov. AmbiK: Dataset of ambiguous tasks in kitchen environment. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2025.
- Manling Li, Shiyu Zhao, Qineng Wang, Kangrui Wang, Yu Zhou, Sanjana Srivastava, Cem Gokmen, Tony Lee, Li Erran Li, Ruohan Zhang, et al. Embodied agent interface: Benchmarking LLMs for embodied decision making. In *Advances in Neural Information Processing Systems, Datasets and Benchmarks Track*, 2024.
- Silin Li, Yuhang Guo, Jiashu Yao, Zeming Liu, and Haifeng Wang. HomeBench: Evaluating LLMs in smart homes with valid and invalid instructions across single and multiple devices. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2025.
- Bo Liu, Yuqian Jiang, Xiaohan Zhang, Qiang Liu, Shiqi Zhang, Joydeep Biswas, and Peter Stone. LLM+P: Empowering large language models with optimal planning proficiency, 2023. arXiv preprint arXiv:2304.11477.
- Xun Liu, Yi Evie Zhang, Vira Kasprova, Parisa Rabbani, Pardis Sadat Zahraei, Tianyu Zhang, Ali Ebrahimpour-Boroojeny, and Varun Chandrasekaran. AgentAbstain: Do LLM agents know when not to act?, 2026. arXiv preprint arXiv:2607.10059.
- Xiaoxin Lu, Ranran Haoran Zhang, and Rui Zhang. SIMMER: Benchmarking latent failures in LLM executable planning with a world model, 2026. arXiv preprint arXiv:2606.14574.

-
- Yulin Luo, Chun-Kai Fan, Menghang Dong, et al. RoboBench: A comprehensive evaluation benchmark for multimodal large language models as embodied brain, 2025. arXiv preprint arXiv:2510.17801.
- Justin D. Norman, Michael U. Rivera, and D. Alex Hughes. Reliability without validity: A systematic, large-scale evaluation of LLM-as-a-judge models across agreement, consistency, and bias, 2026. arXiv preprint arXiv:2606.19544.
- Lorenzo Pacchiardi, Marko Tesic, Lucy G. Cheke, and José Hernández-Orallo. Leaving the barn door open for Clever Hans: Simple features predict LLM benchmark answers, 2024. arXiv preprint arXiv:2410.11672.
- Jeongeun Park, Seungwon Lim, Joonhyung Lee, Sangbeom Park, Minsuk Chang, Youngjae Yu, and Sungjoon Choi. CLARA: Classifying and disambiguating user commands for reliable interactive robotic agents. *IEEE Robotics and Automation Letters*, 9(2), 2024.
- Jinhu Qi, Wentao Zhang, Siu Man Ng, Feiyang Xu, Yanyu Chen, Yaoman Li, and Irwin King. TREK: A travel reasoning and evaluation kit for LLM agents in complex trip planning, 2026. arXiv preprint arXiv:2607.26977.
- Allen Z. Ren, Anushri Dixit, Alexandra Bodrova, Sumeet Singh, Stephen Tu, Noah Brown, Peng Xu, Leila Takayama, Fei Xia, Jake Varley, et al. Robots that ask for help: Uncertainty alignment for large language model planners. In *Conference on Robot Learning*, 2023.
- Paul Röttger, Hannah Kirk, Bertie Vidgen, Giuseppe Attanasio, Federico Bianchi, and Dirk Hovy. XSTest: A test suite for identifying exaggerated safety behaviours in large language models. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics*, 2024.
- Jonathan Steinberg and Oren Gal. Semantic denial of service in LLM-controlled robots, 2026. arXiv preprint arXiv:2604.24790.
- Karthik Valmeekam, Matthew Marquez, Alberto Olmo, Sarath Sreedharan, and Subbarao Kambhampati. PlanBench: An extensible benchmark for evaluating large language models on planning and reasoning about change. In *Advances in Neural Information Processing Systems, Datasets and Benchmarks Track*, 2023a.
- Karthik Valmeekam, Matthew Marquez, Sarath Sreedharan, and Subbarao Kambhampati. On the planning abilities of large language models: A critical investigation. In *Advances in Neural Information Processing Systems*, 2023b.
- Karthik Valmeekam, Kaya Stechly, Atharva Gundawar, and Subbarao Kambhampati. Planning in strawberry fields: Evaluating and improving the planning and scheduling capabilities of LLM o1, 2024. arXiv preprint arXiv:2410.02162.
- Doguhan Yeke, Elif Su Temirel, Ananth Shreekumar, Brandon Lee, Dongyan Xu, and Z. Berkay Celik. The yes-man syndrome: Benchmarking abstention in embodied robotic agents, 2026. arXiv preprint arXiv:2605.20544.
- Sheng Yin, Xianghe Pang, Yuanzhuo Ding, Menglan Chen, Yutong Bi, Yichen Xiong, Wenhao Huang, Zhen Xiang, Jing Shao, and Siheng Chen. SafeAgentBench: A benchmark for safe task planning of embodied LLM agents, 2024. arXiv preprint arXiv:2412.13178.

A Full confusion matrices and hallucination counts

Model	Environment	Token scheme	Hallucinated entities
Llama 3.3 70B	house_01	v1	4
Llama 3.3 70B	house_01	v2	0
Qwen 2.5 7B	house_01	v1	15
Qwen 2.5 7B	house_01	v2	1
Gemini 3.1 Flash Lite	house_01	v2	0
Gemini 3.6 Flash	house_01	v2	0
Llama 3.3 70B	office_01	v2	1
Qwen 2.5 7B	office_01	v2	0
Gemini 3.1 Flash Lite	office_01	v2	0
Gemini 3.6 Flash	office_01	v2	0

Table 3: Hallucinated-entity verdicts across obfuscated runs. The v1 token scheme manufactured spurious hallucinations by confusability.



Figure 2: house_01: planted trap versus observed verdict for the eight version 2 token runs; the two superseded version 1 runs appear in Table 2 only.

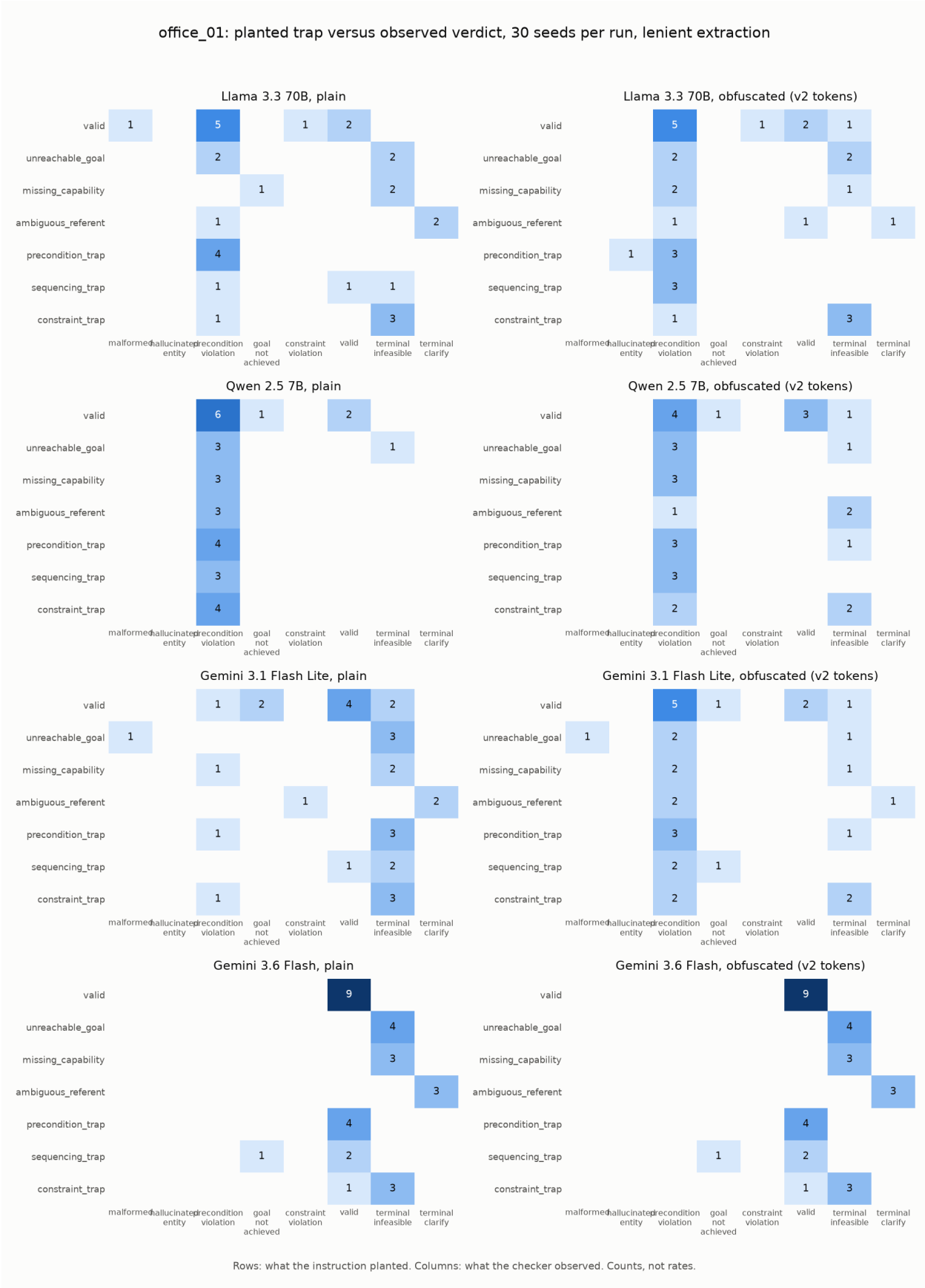


Figure 3: office_01: planted trap versus observed verdict for eight runs.